

Rosen — §4.3: Primes and Greatest Common Divisors

Selected Exercises

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Semester 2, 2026

Exercise 6

How many zeros are there at the end of $100!$?

Exercise 10

Show that if $2^m + 1$ is an odd prime, then $m = 2^n$ for some nonnegative integer n . [*Hint*: First show that the polynomial identity $x^m + 1 = (x^k + 1)(x^{k(t-1)} - x^{k(t-2)} + \cdots - x^k + 1)$ holds, where $m = kt$ and t is odd.]

Exercise 11

Show that $\log_2 3$ is an irrational number. Recall that an irrational number is a real number x that cannot be written as the ratio of two integers.

Exercise 12

Prove that for every positive integer n , there are n consecutive composite integers. [*Hint*: Consider the n consecutive integers starting with $(n + 1)! + 2$.]

Exercise 13

Prove or disprove that there are three consecutive odd positive integers that are primes, that is, odd primes of the form p , $p + 2$, and $p + 4$.

Exercise 18

We call a positive integer *perfect* if it equals the sum of its positive divisors other than itself.

(a) Show that 6 and 28 are perfect.

(b) Show that $2^{p-1}(2^p - 1)$ is a perfect number when $2^p - 1$ is prime.

Exercise 19

Show that if $2^n - 1$ is prime, then n is prime. [*Hint*: Use the identity $2^{ab} - 1 = (2^a - 1) \cdot (2^{a(b-1)} + 2^{a(b-2)} + \cdots + 2^a + 1)$.]

Exercise 21

Recall: the value of the Euler φ -function at the positive integer n is defined to be the number of positive integers less than or equal to n that are relatively prime to n .

Find these values of the Euler φ -function.

(a) $\varphi(4)$.

(b) $\varphi(10)$.

(c) $\varphi(13)$.

Exercise 22

Show that n is prime if and only if $\varphi(n) = n - 1$.

Exercise 31

Show that if a and b are positive integers, then $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$. [*Hint*: Use the prime factorizations of a and b and the formulae for $\gcd(a, b)$ and $\text{lcm}(a, b)$ in terms of these factorizations.]

Exercise 32

Use the Euclidean algorithm to find

(a) $\gcd(1, 5)$.

(b) $\gcd(100, 101)$.

(c) $\gcd(123, 277)$.

(d) $\gcd(1529, 14039)$.

(e) $\gcd(1529, 14038)$.

(f) $\gcd(11111, 111111)$.

Exercise 36

Show that if a and b are both positive integers, then

$$(2^a - 1) \bmod (2^b - 1) = 2^{a \bmod b} - 1.$$

Exercise 37

Use Exercise 36 to show that if a and b are positive integers, then

$$\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a, b)} - 1.$$

[*Hint*: Show that the remainders obtained when the Euclidean algorithm is used to compute $\gcd(2^a - 1, 2^b - 1)$ are of the form $2^r - 1$, where r is a remainder arising when the Euclidean algorithm is used to find $\gcd(a, b)$.]

Exercise 49

Prove that the product of any three consecutive integers is divisible by 6.

Exercise 51

Prove or disprove that $n^2 - 79n + 1601$ is prime whenever n is a positive integer.

Exercise 55

Adapt the proof in the text that there are infinitely many primes to prove that there are infinitely many primes of the form $4k + 3$, where k is a nonnegative integer. [*Hint*: Suppose that there are only finitely many such primes $q_1, q_2, \dots, q_n$, and consider the number $4q_1q_2 \cdots q_n - 1$.]

Exercise 56

Prove that the set of positive rational numbers is countable by setting up a function that assigns to a rational number p/q with $\gcd(p, q) = 1$ the base 11 number formed by the decimal representation of p followed by the base 11 digit A , which corresponds to the decimal number 10, followed by the decimal representation of q .

Exercise 57

Prove that the set of positive rational numbers is countable by showing that the function K is a one-to-one correspondence between the set of positive rational numbers and the set of positive integers if

$$K(m/n) = p_1^{2a_1} p_2^{2a_2} \cdots p_s^{2a_s} q_1^{2b_1-1} q_2^{2b_2-1} \cdots q_t^{2b_t-1},$$

where $\gcd(m, n) = 1$ and the prime-power factorizations of m and n are $m = p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}$ and $n = q_1^{b_1} q_2^{b_2} \cdots q_t^{b_t}$.